

WASP-12b

R-1968

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_151122 · created 2026-08-06T23:37:11+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457347.76062 +/- 0.00022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1253 +/- 0.0071
Transit depth (Rp/Rs)^2	1.57 +/- 0.18 [%]
Orbital Inclination (inc)	81.98 +/- 1.81
Airmass coefficient 1 (a1)	1.1263 +/- 0.0093
Airmass coefficient 2 (a2)	-0.109 +/- 0.0027
Scatter in the residuals of the lightcurve fit is	0.82 %
Best Comparison Star	None
Optimal Aperture	4.27
Optimal Annulus	14.26
Transit Duration (day)	0.1181 +/- 0.0025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1253 $\pm$ 0.0071 vs 0.1178 (deviation 0.91 σ)
Duration fit vs published	0.1181 d vs 0.125 d (deviation 2.39 σ)
Mid-time O-C	-1571.2 min
Depth SNR	15.3
Residual scatter	0.82 %

Light curve

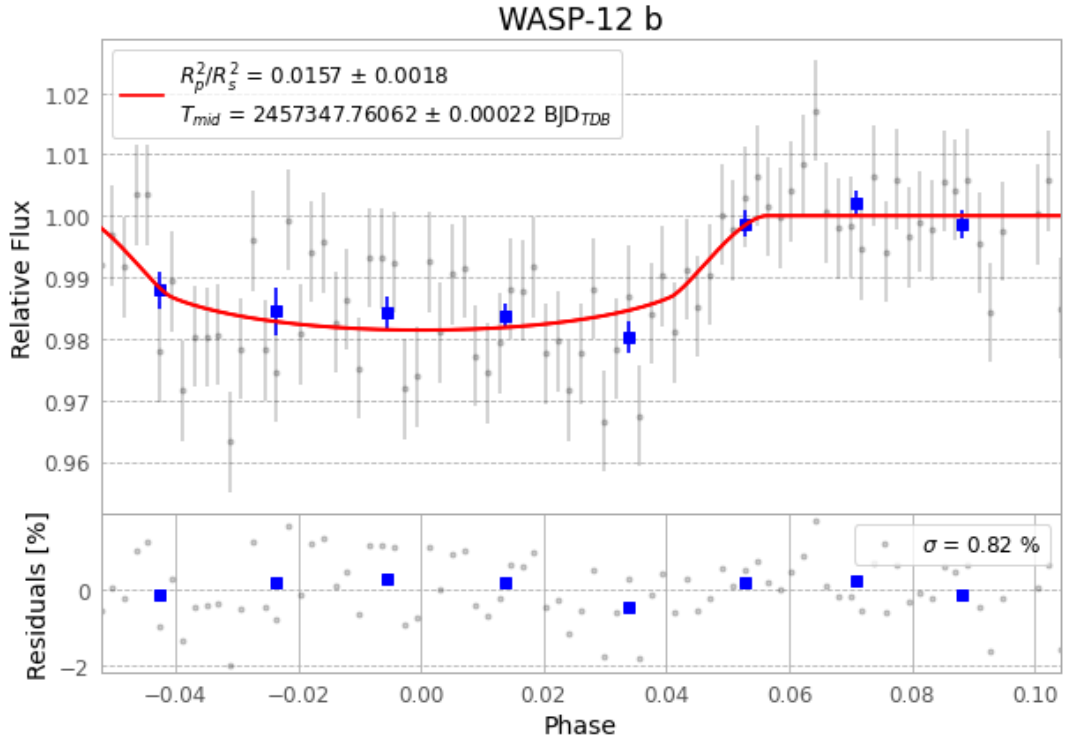


Figure 1 — FinalLightCurve_WASP-12 b_2015-11-20.png (SHA-256 e75c2f8f17de3e74...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [199, 161], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0a71ac5d4bcff81f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

83 FITS frames (55 MB), WASP-12151121043912.FITS → WASP-12151121085112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-151122.log (b5135a3aea74...), FinalLightCurve_WASP-12 b_2015-11-20.png (e75c2f8f17de...), FinalLightCurve_WASP-12 b_2015-11-20.csv (3b081370d8ad...)

Compute resources

Wall time 132.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 23:37Z.